

Exp:4 Best First Search

Program:

```
import heapq

def best_first_search(graph, heuristic, start, goal):
    priority_queue = [(heuristic[start], start)]
    visited = set()

    while priority_queue:
        h, node = heapq.heappop(priority_queue)
        if node in visited:
            continue
        print(node, end=" ")
        visited.add(node)
        if node == goal:
            print("\nGoal Reached!")
            return
        for neighbour in graph[node]:
            if neighbour not in visited:
                heapq.heappush(priority_queue, (heuristic[neighbour], neighbour))
    print("\nGoal Not Found!")

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['G'],
    'F': [],
    'G': []
}

heuristic = {
```

```

'A': 6,
'B': 4,
'C': 5,
'D': 3,
'E': 2,
'F': 4,
'G': 0
}

best_first_search(graph, heuristic, 'A', 'G')

```

Output:

```

*exp 4 program.py - C:/Users/Sakshi/OneDrive/Documents/MLA01 - Lab Exp/Exp - 4/exp 4 program.py (3.13.5)*
File Edit Format Run Options Window Help

import heapq
# Best First Search Algorithm
def best_first_search(graph, heuristic, start, goal):
    priority_queue = [(heuristic[start], start)]
    visited = set()

    while priority_queue:
        h, node = heapq.heappop(priority_queue)

        if node in visited:
            continue

        print(node, end=" ")
        visited.add(node)

        if node == goal:
            print("\nGoal Reached!")
            return

        for neighbour in graph[node]:
            if neighbour not in visited:
                heapq.heappush(priority_queue, (heuristic[neighbour], neighbour))

    print("\nGoal Not Found!")
# Graph Representation
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['G'],
    'F': [],
    'G': []
}
# Heuristic Values
heuristic = {
    'A': 6,
    'B': 4,
    'C': 5,
    'D': 3,
    'E': 2,
    'F': 4,
    'G': 0
}
best_first_search(graph, heuristic, 'A', 'G')
Ln 2 Col: 0

```

```

IDLE Shell 3.13.5
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Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/Sakshi/OneDrive/Documents/MLA01 - Lab Exp/Exp - 4/exp 4 program.py
A B E G
Goal Reached!
>>>
Ln 7 Col: 0

```